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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/247,091	03/29/2023	YUTARO YAMASHITA	SYP335917US01	6920

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EXAMINER	
BROWN JR, HOWARD D	

ART UNIT	PAPER NUMBER
2488	

NOTIFICATION DATE	DELIVERY MODE
12/04/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/247,091	Applicant(s) YAMASHITA et al.	
	Examiner HOWARD D BROWN JR	Art Unit 2488	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 11/08/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,3-5 and 7-14 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,3-5 and 7-14 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/29/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. This Office Action is sent in response to Applicant's communication received on 11/08/2024 for application number 18/247,091. The Office herby acknowledges receipt of the following and placed of record in file: Specification, Drawings, Abstract, Oath/Declaration, and claims.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Applicant(s) Response to Official Action

3. The response filed 11/08/2024 has been entered and made off record.

Response to Arguments

4. Presented arguments have been fully considered, but are rendered moot in view of the new ground(s) of rejection necessitated by amendment(s) initiated by the applicant(s). Examiner's response to the presented arguments follows below.

5. Claim 7 has been amended, as set forth above. Accordingly, the Applicant respectfully requests that the rejection of claim 7 under 35 U.S.C. § 112, second paragraph be withdrawn. Examiner Agreed.

6. **Claims 1, 2, 4-6, and 8** were rejected under 35 U.S.C. § 103 as being obvious over U.S. 2022/0089188 (hereinafter, "Tsugawa'") in view of U.S. 2008/0230681 (hereinafter, "Yoshizu").

The Applicant submits that Tsugawa does not teach, suggest, or render obvious at least, for example, the features of **"...a mirror configured to deflect the infrared light projected from the infrared light source in a direction along an optical axis of the camera unit, wherein the infrared light passes through the transparent material portion based on the deflection,"** as recited in amended independent claim 1.. ***To the examiner the band pass filter disposed in the internal camera for infrared light is equivalent to a mirror that deflects infrared light and a visible light cut filter removing light of the red wavelength component emitted from the near infrared LEDs is provided in front of the projector.|0027].***

7. **Claims 7 and 9-14** were rejected as being obvious over Tsugawa in view of Yoshizu and U.S. 11,287,784 (hereinafter, "Picardi'). The Applicant respectfully traverses the above rejections at least for the reasons previously set forth during prosecution and at least based on the following remarks. The Applicant submits that the combination of Tsugawa, Yoshizu, and Picardi does not teach, suggest, or render obvious at least, for example, the features of **"turn on an infrared light source included in the in-vehicle camera based on a recognition result of an image obtained by the in-vehicle camera,"** as recited in amended independent claim 7. ***To the examiner the camera 530 may be triggered by several different types of techniques (e.g. motion detected, light controlled, preset, Manual) |Col 23 Lines 44-59].***

Claim Rejections - 35 USC § 103

8. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

9. Claim(s) 1-2, and 4-5, 8 is/are rejected under 35 U.S.C. 103 as being unpatentable over Tsugawa [US Pub. No.: 2022/0089188 A1] in view of Yoshizu et al., [US Pub. No.: 2008/0230681 A1 A1].

Re. Claim 1, Tsugawa [US Pub. No.: 2022/0089188 A1] discloses:

An in-vehicle camera ***[internal vehicle camera 3| Fig. 1, 0027]***, comprising: an infrared light source configured to project infrared light ***[camera 3 comprises a projector comprised of two near infrared LEDs placed at the both sides of the camera |0027]***;

a camera unit that includes:

lens unit;

and an imaging device unit wherein the camera unit has ***[The internal vehicle camera 3 includes a lens and an image capturing element and is for example a CMOS (complementary metal oxide semiconductor) camera or a CCD (charge coupled device) camera |0027]*** a sensitivity to a wavelength of the infrared light ***[a band pass filter removing light of wavelength components other than near infrared light is provided inside the camera |0027]***;

and a board wherein the infrared light source and the camera unit are on the board ***[The projector is comprised of LEDs (light emitting diodes). For example, it is comprised of two near infrared LEDs placed at the both sides of the camera |0027]***.

and a mirror configured to deflect the infrared light projected from the infrared light source [***a visible light cut filter removing light of the red wavelength component emitted from the near infrared LEDs is provided in front of the projector.|0027]*** in a direction along an optical axis of the camera unit, wherein the infrared light passes through the transparent material portion based on the deflection ***[a band pass filter removing light of wavelength components other than near infrared light is provided inside the camera, while a visible light cut filter removing light of the red wavelength component emitted from the near infrared LEDs is provided in front of the projector. To the examiner the band pass filter disposed in the internal camera for infrared light is equivalent to a mirror that deflects infrared light and a visible light cut filter removing light of the red wavelength component emitted from the near infrared LEDs is provided in front of the projector.|0027]***.

In the same field of endeavor discloses Yoshizu et al., [US Pub. No.: 2008/0230681 A1] discloses:

a board ***[a camera board 20 |0051,0067]*** a casing that incorporates the board ***[outer case 74 |Fig. 1, 0063]***, wherein the casing includes a transparent material portion that allows the infrared light to pass through ***[Fig.1 Shows a case for housing camera components… the case is used to capture images with in a low lighting environment… it is obvious to have the case be made up of a transparent material to capture images outside of the case |Fig.1 el 74, 0028, 0063, 0073]***;

Therefore, it would have been obvious to one with ordinary skill in the art before the effective filing date of the claimed invention to combine Tsugawa with Yoshizu to have camera housing

base for mounting camera and components together at position to carryout imaging in an in-vehicle environment [0060].

Re. Claim 4, The combination of Tsugawa and Ventimiglia does not disclose:

> The in-vehicle camera according to claim 1, wherein the camera unit includes a light shielding unit configured to shield an imaging surface of the imaging device unit from the infrared light of the infrared light source.

In the same field endeavor Yoshizu discloses:

> The in-vehicle camera according to claim 1, wherein the camera unit includes a light shielding unit configured to shield an imaging surface of the imaging device unit from the infrared light of the infrared light source. ***[Shielding member 78 which is opaque to light… which is substantially perpendicular, blocks incidence on the image capture unit 34, 30 of light, either directly incident from the light-guiding member 10, or reflected by the lower face of the visible light filter 76|t0106].***

Therefore, it would have been obvious to one with ordinary skill in the art before the effective filing date of the claimed invention to combine the internal camera system of Tsugawa with the Yoshizu's camera housing for mounting camera and components together at position to capture images ***[0060]*** .

Re. Claim 5, the rejection of claim 1 is incorporated herein.

Tsugawa et al. meets the claim limitations, as follows:

The in-vehicle camera according to claim 1, wherein the infrared light source includes a plurality of infrared projectors, and the plurality of infrared projectors has a common directivity or

different directivities ***[two infrared LED's for projecting light from both sides of the camera which is interpreted as being equivalent to a different directivity |0027].***

Re. Claim 8, the rejection of claim 1 is incorporated herein.

Tsugawa et al. meets the claim limitations, as follows:

The control apparatus for the in-vehicle camera according to claim 7, wherein the controller is further configured to determine to perform a left turn or a right turn at an intersection based on the recognition result of the image obtained by the in-vehicle camera ***[the processor 83 has a viewpoint detecting part 84, a judging part 85. and a vehicle control part 86 for changing lanes based on captured image wherein the change of lanes is equivalent to turning left or right (e.g. The angles of the facial direction or the line of sight corresponding to the front, the direction of the lane change (right or left), and the direction of the room mirror are determined in advance.) |Fig. 3, 0047, 0053].*** The prior art do not explicitly teach the limitations being implemented at an intersection. It would have been obvious....to assist and allow an operator to safely navigate through an interstation while making turns.

9. Claim(s) 3 is/are rejected under 35 U.S.C. 103 as being unpatentable over Tsugawa in view of Yoshizu in further view of Lu et al., [US Pub. No.: 2012/0019940 A1].

Re. Claim 3, The combination of Tsugawa and Yohizu does not distinctly disclose:

> The in-vehicle camera according to claim 1, wherein the transparent material portion is bonded to the casing by laser welding.

In The same field of endeavor Lu et al., [US Pub. No.: 2012/0019940 A1]discloses:

The in-vehicle camera according to claim 2, wherein the transparent material portion is bonded to the casing based on laser welding ***[Camera housing achieved through laser welding |Fig. Fig.10, 0062]***.

Therefore, it would have been obvious to one with ordinary skill in the art before the effective filing date of the claimed invention to combine Tsugawa and Yoshizu with the vehicular camera and lens assembly of Lu to have a plastic injection molding for lower cost and ease of attaching the back housing 132 to the integrated piece 150 by laser welding. **[0047]**.

10. Claim(s) 7,9-14 is/are rejected under 35 U.S.C. 103 as being unpatentable over Tsugawa in view of Yoshizu in further view of Kimura [US Pub. No.: 2019/0297241 A1].

Re. Claim 7, the rejection of claim 1 is incorporated herein.

Tsugawa et al. meets the claim limitations, as follows:

A control apparatus for an in-vehicle camera ***[internal vehicle camera 3| Fig. 1, 0027]***, ***[The internal vehicle camera 3 includes a lens and an image capturing element and is for example a CMOS (complementary metal oxide semiconductor) camera or a CCD (charge coupled device) camera |0027], [camera 3 is sensitive to infrared wavelength | 0027]***, wherein the in-vehicle camera is on a first vehicle, and the infrared light source and a camera unit are on a board included in the in-vehicle camera ***[camera 3 comprises a projector comprised of two near infrared LEDs placed at the both sides of the camera and is mounted on the surface in a vehicle |Fig.2 0027]***.

Tsugawa does not distinctly disclose:

the control apparatus comprising a controller configured to turn on an infrared light source included in the in-vehicle camera based on a recognition result of an image obtained by the in-vehicle camera.

In the same field of endeavor Picardi discloses

the control apparatus comprising a controller that controls ON of the infrared light source on a basis of a recognition result of an image obtained by the in-vehicle camera ***[The camera 530 may be triggered by several different types of techniques (e.g. motion detected, light controlled, preset, Manual) |Col 23 Lines 44-59]***.

Therefore, it would have been obvious to one with ordinary skill in the art before the effective filing date of the claimed invention to combine Tsugawa with Picardi to have the vehicle monitoring system of configured to trigger camera 530 based low light environment and or motion detected ***[Col 23 Lines 44-59]***.

Re. Claim 9, The combination of Tsugawa and Yoshizu, specifically Tsugawa discloses:

and when based on a determination that overtake of a second vehicle by the first vehicle is planned, and the determination that the overtake is planned is performed based on an internal state of the first vehicle ***[the viewpoint detecting part 84 may detect the angle of the facial orientation or the line of sight… based on detected viewpoint controlling the vehicle |0041]***.

In the same field of endeavor Picardi discloses:

The control apparatus for an in-vehicle camera according to claim 7, wherein the controller is further configured to turn on the infrared light source of the in-vehicle camera based on a ***[The camera 530 may be triggered by several different types of***

techniques (e.g. motion detected, light controlled, preset, Manual) |Col 23 Lines 44-59] a surrounding brightness does not satisfy a condition ***[the camera 530 triggers integrated or external illuminators (e.g., Infra-Red, Z-wave controlled "white" lights, lights controlled by the module 522, etc.) to improve image quality when the scene is dark.|Col 23 Lines 60-65]***

Therefore, it would have been obvious to one with ordinary skill in the art before the effective filing date of the claimed invention to combine Tsugawa with Yoshizu to have camera housing base for mounting camera and components together at position to carryout imaging [0060] along with the vehicle monitoring system of configured to trigger camera 530 based low light environment and or motion detected ***[Col 23 Lines 44-59]***.

Re. Claim 10, the rejection of claim 7 is incorporated herein.

Picardi et al. meets the claim limitations, as follows:

The control apparatus for the in-vehicle camera according to claim 7, wherein the controller is further configured to turn on the infrared light source of the in-vehicle camera based on a determination that a second vehicle turns on a direction indicator and the second vehicle follows the first vehicle. ***[The camera 530 may be triggered by several different types of techniques (e.g. motion detected, light controlled, preset, Manual) |Col 23 Lines 44-59]***

In the same field of endeavor Tsugawa discloses:

when it is determined that a following vehicle turns on a direction indicator ***[the driver instructs a lane change by entering an instruction into the HMI 7 manually or by***

voice, operation of a turn signal lever, etc. It is judged based on the detection signals of these whether the driver has instructed a lane change. |0054].

Therefore, it would have been obvious to one with ordinary skill in the art before the effective filing date of the claimed invention to combine Tsugawa with Yoshizu to have camera housing base for mounting camera and components together at position to carryout imaging [0060] along with the vehicle monitoring system of configured to trigger camera 530 based low light environment and or motion detected ***[Col 23 Lines 44-59]***.

Re. Claim 11, the rejection of claim 7 is incorporated herein.

Picardi et al. meets the claim limitations, as follows:

> The control apparatus for an in-vehicle camera according to claim 7, wherein the controller is further configured to turn on the infrared light source of the in-vehicle camera based on a determination that a second vehicle approaches the first vehicle within a specified distance behind the first vehicle, ***[The camera 530 may be triggered by several different types of techniques (e.g. motion detected, light controlled, preset, Manual) |Col 23 Lines 44-59]***

In the dame field of endeavor Tsugawa discloses:

> the second vehicle approaches the first vehicle at a relative speed which increases, and the second vehicle is on a second lane different from a first lane of the first vehicle ***[The vehicle control part 86, for example, judges that a lane change is necessary for passing when a relative speed of a preceding vehicle with respect to the vehicle 10 is equal to or less than a predetermined value |0053]***.

Therefore, it would have been obvious to one with ordinary skill in the art before the effective filing date of the claimed invention to combine Tsugawa with Yoshizu to have camera housing base for mounting camera and components together at position to carryout imaging [0060] along with the vehicle monitoring system of configured to trigger camera 530 based low light environment and or motion detected ***[Col 23 Lines 44-59]***.

Re. Claim 12, the rejection of claim 7 is incorporated herein.

Picardi et al. meets the claim limitations, as follows:

> The control apparatus for an in-vehicle camera according to claim 7, wherein the controller is configured to set the infrared light source of the in-vehicle camera based on a determination that a second vehicle approaches the first vehicle within a specified distance ahead of the first vehicle, ***[The camera 530 may be triggered by several different types of techniques (e.g. motion detected, light controlled, preset, Manual) |Col 23 Lines 44-59]***

In the dame field of endeavor Tsugawa discloses:

> the second vehicle approaches the first vehicle at a relative speed which decreases and the second vehicle is a second lane different from a first lane of the first vehicle. ***[The vehicle control part 86, for example, judges that a lane change is necessary for passing when a relative speed of a preceding vehicle with respect to the vehicle 10 is equal to or less than a predetermined value.|0053]***.

Therefore, it would have been obvious to one with ordinary skill in the art before the effective filing date of the claimed invention to combine Tsugawa with Yoshizu to have camera housing base for mounting camera and components together at position to carryout imaging [0060] along

with the vehicle monitoring system of configured to trigger camera 530 based low light environment and or motion detected ***[Col 23 Lines 44-59]***.

Re. Claim 13, This claim is interpreted and rejected for the same reason set forth in claim 7.

Re. Claim 14, This claim is interpreted and rejected for the same reason set forth in claim 7 and 13.

Conclusion

THIS ACTION IS MADE FINAL. Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HOWARD D BROWN JR whose telephone number is (571)272-4371. The examiner can normally be reached Monday - Friday 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is

encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Sathyanarayanan Perungavoor can be reached on 5712727455. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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